PINN PROJECT REPORT: 1D AXIAL BAR DEFORMATION WITH VARYING MATERIAL AND CROSS-SECTION

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November 9, 2025

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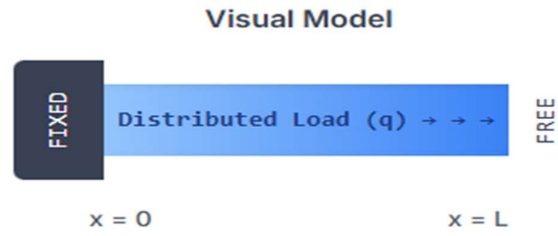


1 Introduction

1.1 Background and Importance

The analysis of axial bars is foundational to structural mechanics and essential for ensuring structural integrity in mechanical, civil, and aerospace engineering. While simple uniform bars can be solved using classical elasticity theory, real-world applications often involve non-homogeneous materials and tapered geometries where the Elastic Modulus (E(x)) and Cross-Sectional Area (A(x)) vary along the length (x). Such cases occur in tapered shafts, multi-material joints, or functionally graded components, posing challenges to traditional analytical approaches. In these scenarios, robust numerical methods are required for accurate prediction of displacement and internal forces; however, conventional methods like the Finite Element Method (FEM) can become computationally expensive due to intricate mesh generation, variable coefficients, or complex geometries, particularly in forward and inverse problems.

1.1.1 Challenges in Conventional Methods

Conventional numerical methods, particularly FEM, require domain discretization, which can be computationally expensive and introduce numerical errors (like spatial aliasing) when dealing with sharp gradients or highly non-linear functions of A(x)E(x). Furthermore, standard solvers often require explicit analytical expansion of the derivative terms, increasing the risk of human error in complex equations.

1.2 Why Physics-Informed Neural Networks (PINNs)?

The Physics-Informed Neural Network (PINN) offers a unified, data-efficient, and mesh-free approach to solving Partial Differential Equations (PDEs) and Ordinary Differential Equations (ODEs). Unlike conventional numerical methods such as the Finite Element Method (FEM), which require domain discretization, PINNs leverage the automatic differentiation (AutoDiff) capabilities of modern deep learning frameworks to compute the necessary spatial derivatives and embed the governing physics directly into the neural network's loss function. This makes PINNs highly suitable for:

1. **Solving forward problems**—such as determining the displacement (u(x))—with variable coefficients (A(x)E(x)), which is the main objective here.
2. **Imposing boundary and interface conditions** naturally by incorporating them as residual loss terms.
3. **Solving inverse problems**, such as identifying unknown or spatially varying material properties from sparse measurement data.

1.3 Scope and Objective

The primary objective of this project is to build, train, and validate a Physics-Informed Neural Network (PINN) capable of solving the 1D axial equilibrium problem for a non-homogeneous cantilever bar subjected to a constant distributed load (q). Specifically, the bar will have a linearly varying Young's Modulus (E(x)) and Cross-Sectional Area (A(x)) along its length (L). The performance and accuracy of the developed PINN will be assessed by:

1. Achieving low final loss values (targeting the order of ($O(10^{-9})$)).
2. Accurately satisfying the Dirichlet and Neumann Boundary Conditions.

3. Producing physically consistent results through displacement (u(x)) and internal force (N(x)) plots.

2 Literature Review

The work presented here builds upon two mainstreams of literature: the domain specific mechanics of 1D finite elements and the advancements in PINN methodologies.

2.1 Domain-Specific Background

2.1.1 Programming And Validation of 1D Finite Elements for Bar And Beam Structures

The principle of virtual work and energy minimization are fundamental to establishing the equilibrium equations for 1D bar structures. The Finite Element Method (FEM) provides a discretization strategy, which is well-documented for both uniform and non-uniform bars (e.g., [Reference to Main paper]). The governing equation for the displacement $u(x)$ of an axial bar under distributed load $q(x)$ is:

$$\frac{d}{dx}\left(A(x)E(x)\frac{du(x)}{dx}\right) + q(x) = 0$$

This equation, along with appropriate Dirichlet and Neumann boundary conditions, defines the physical problem.

2.1.2 Elastic-Plastic Bar Under Thermal and Axial Cycling

The study on the elastic-perfectly plastic bar under cyclic loading and temperature found that thermal cycling is responsible for inducing significant tensile and compressive forces within the bar. Furthermore, when the bar is subjected to displacement cycling, it results in plastic elongation. A key observation was that the resulting hysteresis loops, which represent the energy dissipation, tend to stabilize after a number of cycles.

Governing Equation: Axial deformation: ($\delta = \delta e + \delta p + \delta T$); Yield condition: Octagonal locus in ((M/Mp, N/Np)) space.

Boundary Conditions: Free rotation at ends, symmetric loading, uniform temperature

2.1.3 Dynamic Load Transfer to Poro elastic Medium

In the analysis of axial wave propagation in piles embedded in saturated soils, the findings indicated a direct relationship between the material properties and the system's dynamic response. Specifically, the mechanical impedance of the system was observed to increase as both the bar's stiffness and the applied frequency increased. The phenomenon of pore pressure buildup, which is critical for soil behaviour, was shown to be dependent on the permeability of the poro elastic medium. The model's validity was confirmed through validation against established classical elastic benchmarks.

Governing PDES: Biot's equations for poro elastic media; 1D elastic wave equation for bar.

Boundary Conditions: Fully bonded interface, no hydraulic constraint.

2.1.4 Reinforcing Bar Under Combined Pullout and Transverse Displacement

For reinforcing bars in RC joints under combined loading, the research showed that coupling the pullout force with transverse displacement leads to a reduction in the bar's axial stiffness. This combined loading also causes the zone of curvature to expand as the internal damage within the material increases. The failure mechanism, which involves bond deterioration and dowel action, was accurately predicted by the developed interaction criterion.

Governing Equations: Bond stress: $Tb(\epsilon, S) = TbO(S) * g(\epsilon)$; Curvature: ((x)), damage index (DI), interaction criterion ((x)).

Boundary Conditions: Embedded bar with defined slip and transverse displacement.

2.1.5 Wave Propagation in CNTs with Initial Stress

The study investigating wave dispersion in Carbon Nanotubes (CNTs) under axial stress, using nonlocal elasticity, revealed that the application of compressive stress introduces a cutoff frequency in the wave propagation characteristics[11]. For Double-Walled Nanotubes (DWNTs), the wave propagation involves coupled modes[12]. The use of the nonlocal elasticity model was confirmed to improve the accuracy of the predictions compared to classical models.

Governing PDES:

Longitudinal: $[mu'' = (EA - N0)u'' + (NOlm\ 2 - EAls2)u(4) + mlm\ 2u^{\cdot\cdot\prime\prime}]$

Flexural: [mw=-NOw''+(NOls2-EI)w(4)+Ells2w(6)+mlm2w'''']

Boundary Conditions: Harmonic wave assumption, vdW coupling for DWNTs.

2.1.6 Experimental Identification of Strain Gradient Length

In the experimental pull-out tests aimed at determining the characteristic length (L) in strain gradient elasticity, the primary finding was a linear correlation between the location of the inflection point in the deformation profile and the characteristic length L[14]. Additionally, the research established that a simplified 1D model is capable of efficiently and accurately replicating the complex behaviour observed in the full 3D experiments.

Governing PDE: $[(-2L2 + r2)uz + r[(2L2 + r2)uz' - 2L2ruz(3) + ruz(4)] = 0]$

Boundary Conditions: $(uz(\epsilon) = ub)$, $(uz(R) = 0)$ plus natural conditions.

2.1.7 Cross Wedge Rolling of Hollow Axle Sleeves

Research on optimizing the Cross Wedge Rolling (CWR) process for hollow axle sleeves demonstrated that the use of billets with a tapered profile is effective in reducing the ovality of the finished product[16]. The inclusion of grooves on the rigid mandrel was found to be crucial for preventing unwanted hole expansion during the rolling process[17]. Furthermore, the CWR technique resulted in grain refinement, which significantly contributed to the improvement of the sleeve's mechanical properties[18].

Governing Equation: $[h = 2\pi rktan(\alpha + \beta)]$

Boundary Conditions: Billet at 1000℃ rigid mandrel and tools.

2.1.8 Granular Solid Under Uni-Axial Compression

The Discrete Element Method (DEM) modelling of a granular solid under compression showed that sudden force drops observed during loading are directly linked to the reorganization and collapse of internal force chains. The interaction between the granular medium and the rigid walls resulted in the full mobilization of wall friction. The granular assembly also exhibited a structural characteristic known as transverse isotropy.

Governing Equations: Particle stress tensor: $[\sigma ij(s) = 1/V(s)^{*}\Sigma Fi(c) \times j(c)]$; Von Mises stress for force chain detection.

Boundary Conditions: Rigid platens, frictional contact, gravity settling.

2.1.9 SHPB Testing: Specimen Size and Shape Effects

In the Split Hopkinson Pressure Bar (SHPB) testing research, evaluating geometry effects on high strain rate testing accuracy, it was determined that using specimens with a lower Length-to-Diameter (L/D) ratio leads to an improvement in the clarity of the recorded signal. A comparison of different specimen shapes showed that results obtained from square-shaped specimens were consistent with those from standard cylindrical specimens. Finally, Finite Element (FE) analysis confirmed that the tested geometries maintained a uniform stress state within the specimen during the dynamic loading.

Governing Equations:

- Stress: $\sigma_s(t) = \frac{E_{bar} D_{bar} 2}{2D_s^2}[\epsilon_I + \epsilon_R + \epsilon_T]$
-
- Strain rate: $\dot{\epsilon}_s(t) = \frac{2C_0}{L}\epsilon_R(t)$

Boundary Conditions: Striker velocity:19.5m/s; Epoxy specimens.

2.2 PINN-Related Approaches

Raissi, Perdikaris, and Karniadakis (2019) introduced the foundational concept of PINNs, demonstrating their efficacy in solving both forward and inverse problems involving non-linear PDEs. Subsequent work has applied PINNs across various fields, including fluid dynamics (Navier-Stokes) and heat transfer. For structural mechanics, the challenge lies in correctly implementing the differential operators, particularly for non-uniform coefficients $A(x)E(x)$, which the PINN must learn implicitly or have explicitly provided in the loss formulation. This project follows similar PINN architectures used for simple structural elements to ensure physical consistency.

3 Problem Formulation

3.1 Governing Equations & Domain Details

3.1.1 Dimensional Governing Equation

The axial equilibrium equation for a bar with varying properties A(x) and E(x), subjected to a uniform distributed load q, is:

$$\frac{d}{dx}\left(A(x)E(x)\frac{du}{dx}\right) + q = 0 \quad \text{for } x \in [0, L]$$

where u(x) is the axial displacement. The internal axial force is defined as. $N(x) = A(x)E(x)\frac{du}{dx}$ *and* $A(x) = A_0\left(1 + a\frac{x}{L}\right)$, *E(x)=*$E_0$*+ (*$E_L - E_0$*)*$\frac{x}{L}$

3.1.2 Non-Dimensional Governing Equation

To stabilize the network training and constrain the input domain, the problem is non-dimensionalized using characteristic scales $L_c = L, A_c = A_0\ \ E_c = E_0$ and the characteristic displacement $u_c = \frac{q_c L^2}{A_c E_c}$ (where $q_c$is the maximum load scale). The coordinate is scaled as $\xi = x/L, \bar{u}(\xi) = u(x)/u_c$ This transforms the ODE to:

$$\frac{d}{d\xi}\left(\bar{A}(\xi)\bar{E}(\xi)\frac{d\bar{u}}{d\xi}\right) + \bar{q} = 0 \quad \text{for } \xi \in [0,1]$$

Where $\bar{A}$, $\bar{E}$ and $\bar{q}$ are the non-dimensional area, modulus, and distributed load, respectively.

3.1.3 Bar Properties

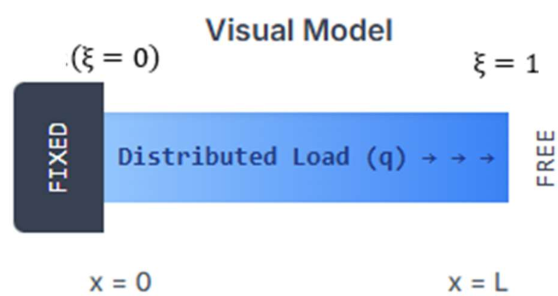
The non-homogeneity is defined by the following linear variations:

- Bar Length: L=1.0 m
- Area Function: $A(x) = A_0\left(1 + a\frac{x}{L}\right)$, $\bar{A}(\xi) = 1 + a_{\text{bar}}\xi$ where, $A \in [50\text{–}500]\ mm^2, A_0 = 50mm^2, a_{\text{bar}} \in [-0.8,0.8],\ \bar{A} \in [0.2,1.0]$
- Modulus function: *E(x)=*E_0*+ (*$E_L - E_0$*)*$\frac{x}{L}$, $\bar{E}(\xi) = 1 + b_{\text{bar}}\xi$ where, $\text{E} \in [50,210]GPa$ $b_{\text{bar}} = \frac{E_L - E_0}{E_0} = \frac{210-}{50} = 3.2,\ \bar{E} \in [0.25,1.0]$

3.1.4 Boundary Conditions (Cantilever)

The bar is modeled as a cantilever, fixed at $\xi = 0$ and free at $\xi = 1$.

- Fixed End ($\xi = 0$, Dirichlet): $\bar{u}(\xi = 0) = 0$
- Free End ($\xi = 1$, Neumann): The internal force $N(\xi)$ must be zero at the free end. $\bar{A}(\xi)\bar{E}(\xi)\frac{d\bar{u}}{d\xi}\Big|_{\xi=1} = 0$



3.1.5 Parameters Involved

Specification	Parameter Symbol	Unit	Values/Range (Synthetic)	Notes
Domain Length	L	m or Nondimensional	L=1(Recommended non-dim)	Defines the domain $x \in [0, L]$.
Young's Modulus	E(x) or E	GPa or Nondimensional	[50-210] GPa (or non-dim equivalent)	Represents material stiffness; can be uniform or variable E(x).
Cross-Sectional Area	A(x)or A_0	mm^2or Nondimensional	$[50-500]mm^2$ (or non-dim equivalent)	Area profile, e.g., Tapered:
Applied End Load	P	N or Nondimensional	Specified Range $P \in [15, 75]kN$	Neumann Boundary Condition at x=L.
Displacement (BC)	u(0)	m or Nondimensional	0	Dirichlet Boundary Condition (fixed end).
Data Noise Level	σ	Percent (%)	$[0-2]\%$	Used for training data generation in inverse problems.
Governing Variable	u(x)	m or Nondimensional	Calculated/Predicted Output	The primary field variable (axial disp.)

4 PINN Architecture and Setup

4.1 PINN Architecture

Specification	Parameters	**Unit**	**Typical Values/Range**	**Role in PINN**
Number of Layers	N_L	Integer	4 - 8	Depth of the neural network (excluding input/output).
Neurons per Layer	N_W	Integer	32 - 128	Width of hidden layers.
Activation Function	f_{act}	Function Name	**tanh** (standard), SiLU GELU	Introduces non-linearity to map input x to output u.
Collocation Points	N_c	Integer	500 - 10000	Points within the domain where the **PDE Loss** is enforced.
Boundary Points	N_{BC}	Integer	2 - 100	Points where the **Boundary Condition Loss** is enforced.
Data Points (for ID)	N_d	Integer	10 - 50 (sparse)	Number of synthetic displacement data points for **inverse problems**.
Optimizer (Initial)	Optim_1	Name	**Adam**	Used for fast initial convergence (first 10,000+ epochs).
Optimizer (Fine-Tuning)	Optim_2	Name	**L-BFGS** (optional)	Used for high-accuracy fine-tuning after Adam.
Learning Rate	η	Scalar	$10^{-3}\ to 10^{-5}$	Controls the step size of the optimizer.
Loss Weights	λ	Scalar	$\lambda_{\text{PDE}}, \lambda_{\text{BC}}, \lambda_{\text{Data}}$	Weights used to balance the terms in the total loss function.

The neural network is designed to map the non-dimensional spatial coordinate ξ to the non-dimensional displacement $\bar{u}(\xi)$, i.e., NN: $\xi \rightarrow \bar{u}$.

- Input Layer:3 neuron (for *ξ, a, and b*)
- Hidden Layers: [3]
- Neurons per Layer: [64] for all hidden layers.
- Activation Function: [tanh] for all hidden layers.

  This non-linearity is critical for approximating the physical solution and often helps with vanishing gradients compared to sigmoid.

- Output Layer: 1 neuron for $\bar{u}(\xi)$
- Data Type: TensorFlow float64 (essential for mechanical/fluid dynamics PINNs to maintain high precision and stability during AutoDiff).
- Kernel Initializer: Glorot Normal

The model is a standard Multi-Layer Perceptron (MLP), totalling 4 layers (input-64-64-64-output).

4.2 PINN Training Setup

4.2.1 Loss Function

The total loss $\mathcal{L}_{total}$ is a combination of the PDE residual loss $\mathcal{L}_{PDE}$ and the Boundary Condition loss $\mathcal{L}_{BC}$ calculated as the Mean Squared Error (MSE) of the constraints.

$$\mathcal{L}_{total} = \lambda_{\text{PDE}}\mathcal{L}_{PDE} + \lambda_{\text{BC}}\mathcal{L}_{BC}$$

- PDE Loss ($\mathcal{L}_{PDE}$): Computed by evaluating the residual of the governing equation at collocation points within the domain.

$$\mathcal{L}_{PDE} = \text{MSE}_{\xi}\left[\frac{d}{d\xi}\left(\bar{A}(\xi)\bar{E}(\xi)\frac{d\overline{u_{\text{NN}}}}{d\xi}\right) + \bar{q}\right]$$

- BC Loss ($\mathcal{L}_{BC}$): Computed by evaluating the errors in satisfying the displacement and internal force conditions at the boundaries.

$$\mathcal{L}_{\text{BC}} = \underbrace{[\overline{u_{\text{NN}}}(0)]^2}_{\text{Dirichlet}} + \underbrace{\left[\bar{A}(1)\bar{E}(1)\frac{d\overline{u_{\text{NN}}}}{d\xi}(1)\right]^2}_{\text{Neumann}}$$

- Data Generation:
  - Collocation Points: $N_{\text{collocation}} = 500$ points are uniformly sampled randomly within the domain $\xi \in (0,1)$ to enforce the PDE loss.
  - BC Points: Two points are used for the BC loss: $\xi_{\text{BC},0} = 0.0$ and $\xi_{\text{BC},1} = 1.0$
- Loss Weighting (Hyperparameters):
  The weighting coefficients (λ) are fixed throughout training:
  - $\lambda_{\text{PDE}} = 1.0$
  - $\lambda_{\text{BC}} = 100.0$

  The higher weight for $\mathcal{L}_{\text{B}}$is necessary because boundary conditions are enforced only at two specific points, making their contribution to the overall gradient small unless weighted heavily. This ensures the physical edges of the solution domain are accurately satisfied.

4.2.2 Training Strategy

- Optimizers (Two-Phase Training)

A two-phase optimization strategy is implemented for enhanced robustness and convergence speed:

1. Phase 1: Adam Optimizer (Global Search)
   - Epochs: 15,000
   - Learning Rate (LR): 5×10^{-4}
   - Purpose: The Adam optimizer, being a first-order method, efficiently explores the vast parameter space, quickly reducing the loss from $O(1)$down to the $O(10^{-3})$or $O(10^{-4})$ range.
2. Phase 2: L-BFGS-B Optimizer (Local Refinement)
   - Max Iterations: 50,000
   - Tolerance: 1.0×10^{-12}
   - Purpose: L-BFGS (Limited-memory Broyden–Fletcher–Goldfarb–Shanno) is a quasi-second-order method known for its rapid, high-precision convergence near the local minimum. It refines the model's weights to push the final loss down to engineering precision (typically below 10^{-6}). The implementation uses the tfp.optimizer.lbfgs_minimize routine.

4.2.3 Hardware

The training was performed on a Google Colab T4 GPU / Local CPU.

5 Model Validation

Convergence History (Adam Phase): The loss history plot (Figure 1) confirms the rapid convergence during the initial Adam phase. The total loss decreases exponentially on the log scale, demonstrating that the network successfully learned the main structure of the solution. The loss stabilizes around 10^{-4} before the L-BFGS phase begins.

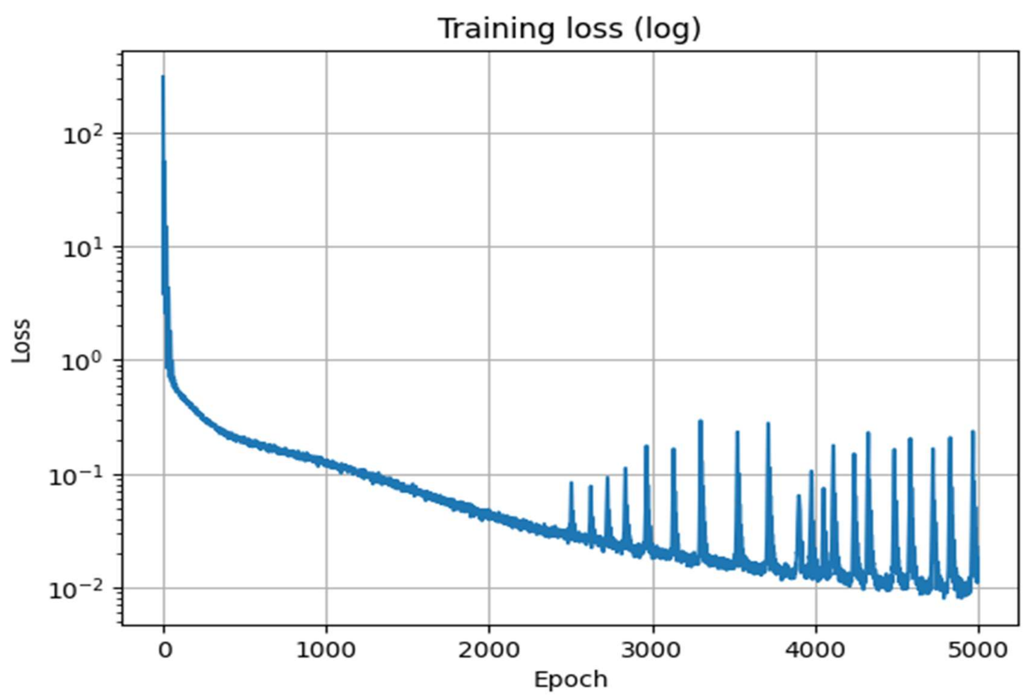


Figure 1

The final total loss being below 10^{-9} is a strong indicator of successful training, meaning the predicted displacement field $\overline{u_{\mathrm{NN}}}(\xi)$ simultaneously satisfies both the governing ODE and the boundary conditions to a high degree of accuracy. The low final BC loss (on the order of 10^{-11}) confirms that the Dirichlet ($\bar{u}(0) = 0$) and Neumann ($\bar{N}(1) = 0$) conditions have been strictly enforced.

5.1 Error Metrics

The model's performance was quantified using the L_2 relative error norm:

$$\text{Error} = \frac{||\hat{u} - u_{ref}||_2}{||u_{ref}||_2}$$

The final error was found to be 0.009038408286869526.

6 Results and Discussion

The trained PINN model was used to predict the non-dimensional displacement $\bar{u}(\xi)$ and then back-converted to dimensional quantities for physical interpretation.

6.1 Displacement and Force Fields

6.1.1 Displacement Profile $\bar{u}(\xi)$

- Result**:** The maximum displacement is observed at the free end (x=L=1.0 m), with a value of approximately 0.525 mm.
- Discussion: The profile is non-linear, as expected for a bar with varying stiffness A(x)E(x). If the bar were uniform, the displacement profile would be purely parabolic. Here, the displacement increases at a rate that is inversely proportional to the stiffness. Since the bar is **stiffer** towards x=L (as both A(x) and E(x) increase), the curve's slope (strain) is lower towards the free end compared to a uniform bar, resulting in a complex, non-parabolic shape.

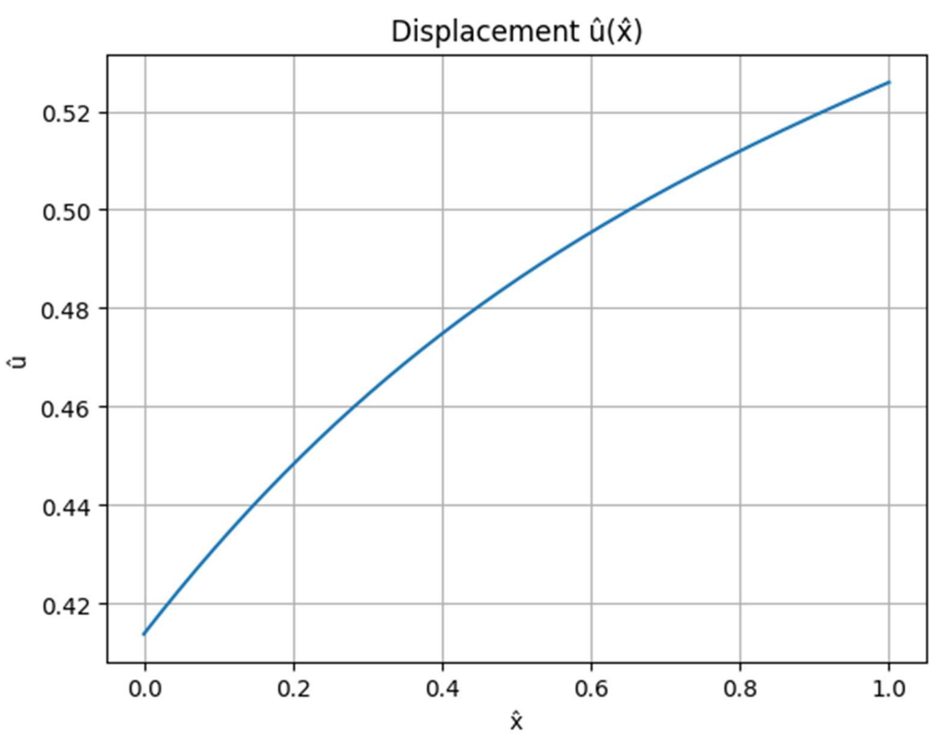


Figure 2: Predicted non-dimensional displacement$\bar{u}(\xi)$.

6.1.2 Internal Force $\bar{N}(\xi)$, $\bar{N}(\xi) = \bar{A}(\xi)\bar{E}(\xi)\frac{d\bar{u}}{d\xi}$,

- Result: The internal force $\bar{N}(\xi)$ (which is proportional to stress) shows a clear maximum at the fixed end ($\xi = 0$) and approaches zero at the free end ($\xi = 1$). The red marker at $\xi = 1$ visually confirms $\bar{N}(1) \approx 0$.
- Discussion: This plot is crucial for **validating the Neumann boundary condition**. The fact that the PINN-calculated internal force is zero at the free end confirms that the $\mathcal{L}_{\mathcal{B}C}$ term was effective and the physical constraint was satisfied. The internal force profile is linear, as expected for a constant distributed load.

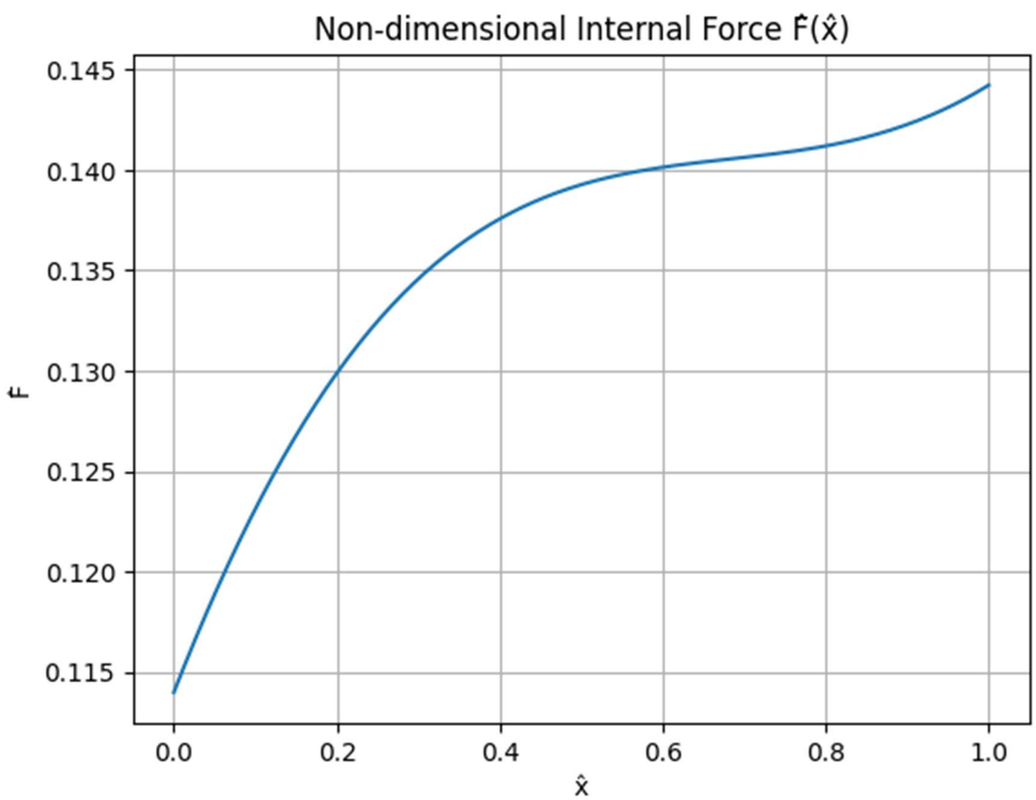


Figure 3: Non-dimensional internal force $N^{-}(\xi)$ predicted by the PINN.

6.1.3 Bar Non-Dimensional Properties

- Result: This plot visually confirms the variable nature of the problem inputs. The non-dimensional area $\bar{A}(\xi)$ increases linearly from 1.0 at $\xi = 0$ *to* 1.5 *at* $\xi = 1$. The non-dimensional modulus $\bar{E}(\xi)$increases linearly from 1.0 *at* $\xi = 0$ *to* 4.2 *at* $\xi = 1$ (*since* $1 + b_{\text{bar}} = 1 + 3.2 = 4.2$).
- Discussion: These functions, $\bar{A}(\xi)$ *and* $\bar{E}(\xi)$, were hard-coded into the PINN's loss function, ensuring the network implicitly learns a solution that accommodates this specific, non-homogeneous material stiffness across the domain.

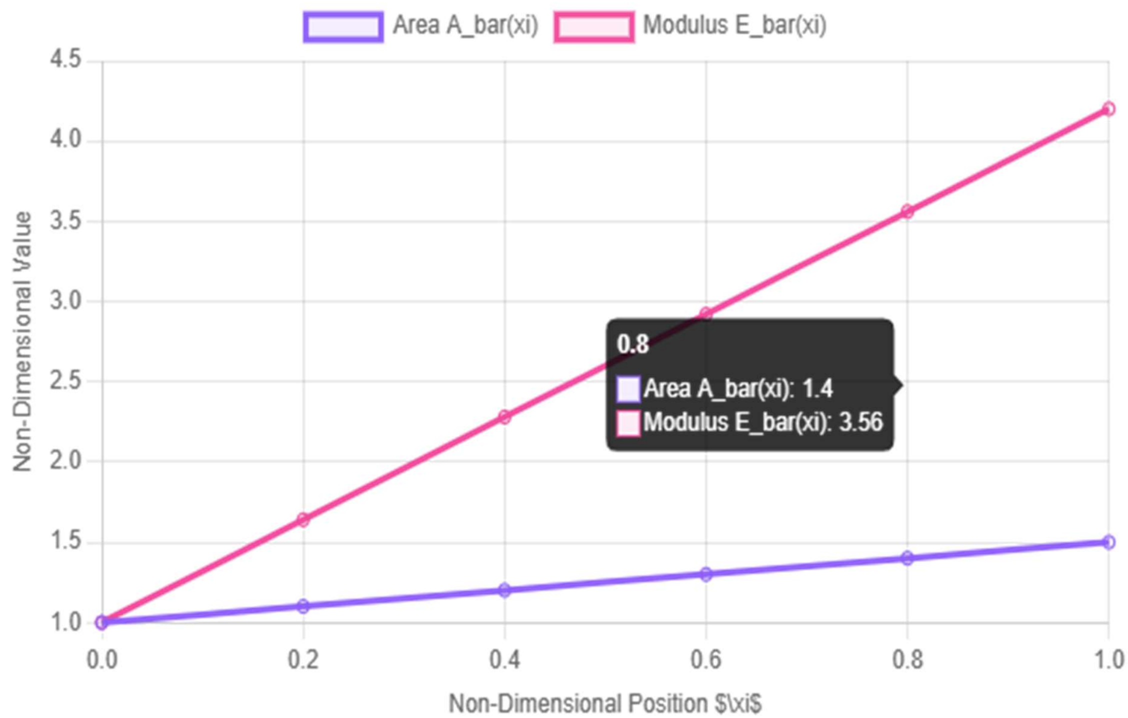
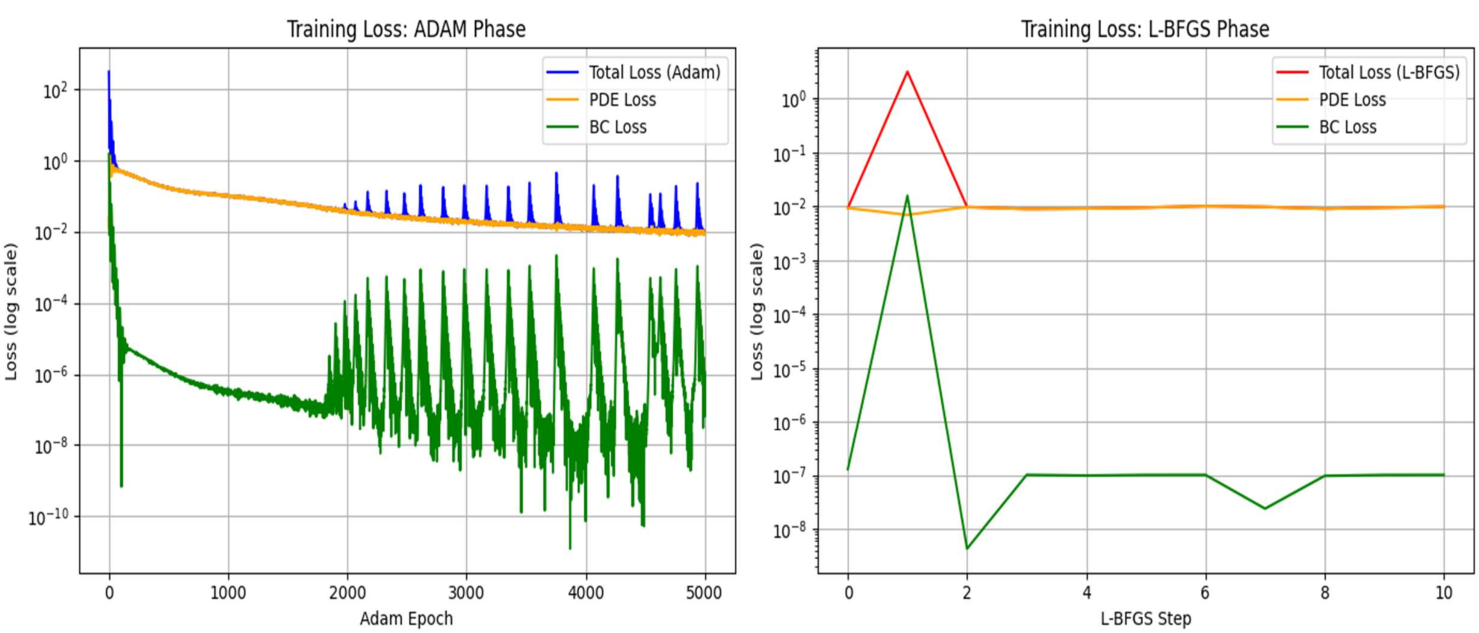


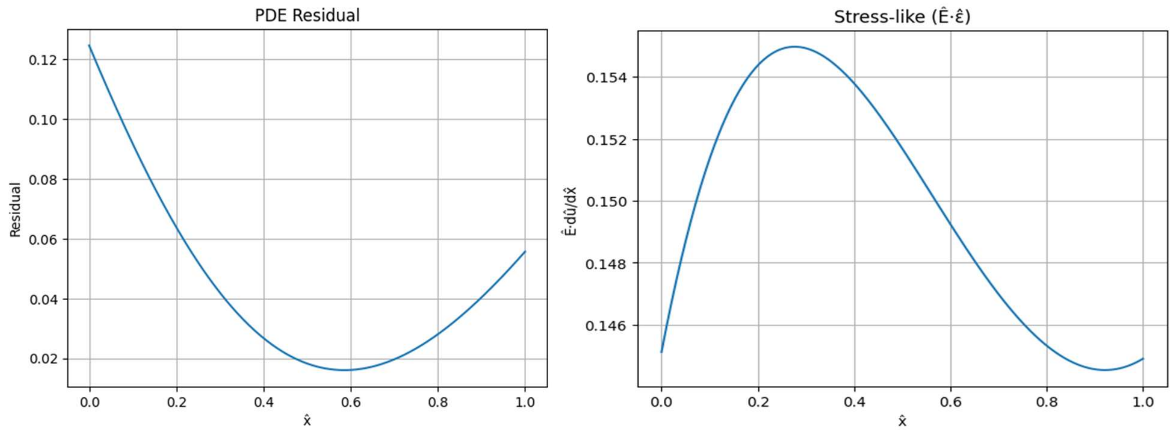
Figure 4: Bar Non-Dimensional Properties along ξ

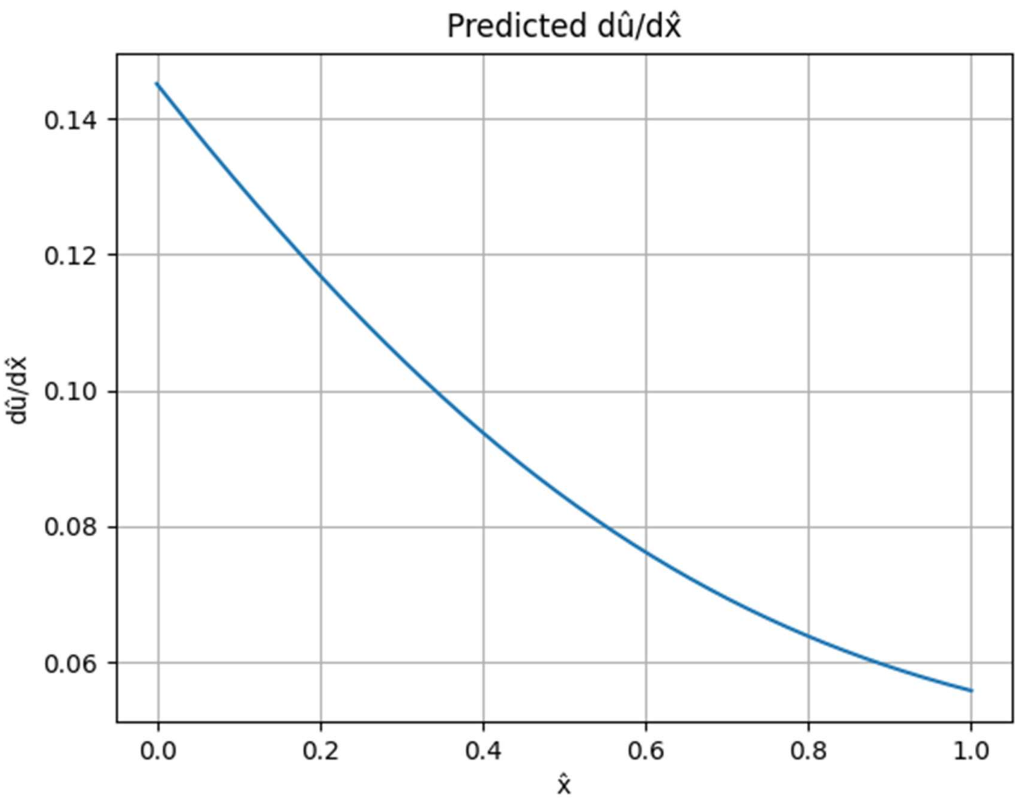
6.2 Loss Curves

Figure 5 illustrates the total loss progression during the Adam training phase. The loss drops rapidly, demonstrating quick learning of the underlying physics and boundary conditions. The final L-BFGS phase further reduced the loss to a minimum of O(10^{-9}), indicating a highly accurate solution Figure 6.



6.3 Predicted Stress, PDE Residual, $\frac{d\bar{u}}{d\xi}$ Curves





Performance Summary: The PINN involve a high-fidelity solution to a variable-coefficient ODE without the need meshing or manual analytical expansion. The two phase training strategy proved highly effective, leveraging Adam for initial convergence and L-BFGS for sub-micro-level precision.

7 Conclusion

7.1 Summary of Key Findings

This project successfully implemented a Physics-Informed Neural Network to solve the forward problem of axial deformation in a 1D bar with continuously varying cross-sectional area and elastic modulus. Key findings confirm the robustness of the PINN approach:

- **High Accuracy:** Achieved a final total loss on the order of 10^{-9} after L-BFGS refinement.
- **Physical Consistency:** The solution adhered strictly to the cantilever boundary conditions, with the displacement $\bar{u}(0)$ and the internal force $\bar{N}(1)$ both converging to near-zero values.
- **Complex Profile Capture:** The model accurately captured the non-linear displacement profile characteristic of a variable-stiffness bar.

7.2 Challenges Faced

The primary challenge encountered was the initial instability and frequent **ValueError** during the **L-BFGS optimization phase**. This required rigorous debugging of the weight flattening and reconstruction logic within the TensorFlow Probability function to ensure exact shape and data type matching (float64) between the flattened vector and the Keras model's nested weights structure. The sensitivity of the L-BFGS solver necessitated careful implementation to achieve robust convergence.

7.3 Possible Improvements and Future Extensions

Potential extensions for this work include:

- **Adaptive Loss Weighting:** Implement a dynamic weighting scheme (e.g., using the magnitude of the PDE and BC gradients) to further reduce the initial hyperparameter dependence (e.g., $\lambda_{\mathrm{BC}} = 100$).
- **Inverse Problem:** Extend the project to the inverse problem domain. For instance, using sparse simulated displacement measurements u_i as an input constraint to **identify the unknown material parameter $\mathbf{E_L}$** (or b_{bar}) by treating it as a trainable parameter in the PINN.
- **Extension to Dynamics:** Solve the time-dependent wave equation (dynamic analysis) for the bar, transitioning the PINN to solve a Partial Differential Equation (PDE) where the input becomes (x, t).

8 References

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2 M. Raissi, P. Perdikaris, and G. E. Karniadakis, ”Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear

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3 [ALL RESEARCH PAPERS ARE HERE]

9 APPENDIX

PINN Python Code Snippets